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Can the Market Add and Subtract? Mispricing in Tech Stock Carve-outs

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Recent equity carve-outs in U.S. technology stocks appear to violate a basic premise of financial theory: identical assets have identical prices. In our 1998–2000 sample, holders of a share of company A are expected to receive x shares of company B, but the price of A is less than x times the price of B. A prominent example involves 3Com and Palm. Arbitrage does not eliminate this blatant mispricing due to short-sale constraints, so that B is overpriced but expensive or impossible to sell short. Evidence from options prices shows that shorting costs are extremely high, eliminating exploitable arbitrage opportunities.

I. Introduction

There are two important implications of the efficient market hypothesis. The first is that it is not easy to earn excess returns. The second is that prices are “correct” in the sense that prices reflect fundamental value. This latter implication is, in many ways, more important than the first. Do asset markets offer rational signals to the economy about where to

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invest real resources? If some firms have stock prices that are far from intrinsic value, then those firms will attract too much or too little capital. While important, this aspect of the efficient market hypothesis is difficult to test because intrinsic values are unobservable. That is why tests of relative valuation, for example, using closed-end funds, are important. The fact that closed-end funds often trade at substantial discounts or premia makes one wonder whether other assets may also be mispriced.

The most basic test of relative valuation is the law of one price: the same asset cannot trade simultaneously at different prices. The law of one price is usually thought to hold nearly exactly in financial markets, where transactions costs are small and competition is fierce. Indeed, the law of one price is in many ways the central precept in financial economics. Our goal in this paper is to investigate violations of the law of one price, cases in which prices are almost certainly wrong in the sense that they are far from the frictionless price. Although the number of cases we examine is small, the violations of the law of one price are large.

The driver of the law of one price in financial markets is arbitrage, defined as the simultaneous buying and selling of the same security for two different prices. The profits from such arbitrage trades give arbitrageurs the incentive to eliminate any violations of the law of one price. Arbitrage is the basis of much of modern financial theory, including the Modigliani-Miller capital structure propositions, the Black-Scholes option pricing formula, and the arbitrage pricing theory.

Do arbitrage trades actually enforce the law of one price? This empirical question is easier to answer than the more general question of whether prices reflect fundamental value. Tests of this more general implication of market efficiency force the investigator to take a stance on defining fundamental value. Fama (1991, p. 1575) describes this difficulty as the "joint-hypothesis" problem: "market efficiency per se is not testable. It must be tested jointly with some model of equilibrium, an asset-pricing model." In contrast, one does not need an asset-pricing model to know that identical assets should have identical prices.

The same difficulty that economists face in trying to test whether asset prices generally reflect intrinsic value is also faced by real-world arbitrageurs looking for mispriced securities. For example, suppose that security A appears to be overpriced relative to security B. Perhaps A is a glamorous growth stock, say a technology stock, and B is a boring value stock, say an oil stock. An arbitrageur could short A and buy B. Unfortunately, this strategy is exposed to "bad-model" risk, another name for the joint-hypothesis problem. Perhaps the arbitrageur has neglected differences in liquidity, risk, or taxes, differences that are properly reflected in the existing prices. In this case, the trade is unlikely to earn excess returns. Researchers have not been able to settle, for

example, whether value stocks are too cheap relative to growth stocks (as argued by De Bondt and Thaler [1985] and Lakonishok, Shleifer, and Vishny [1994]) or just more risky (as favored by Fama and French [1993]).

Another, second, risk for the arbitrageur is fundamental risk. An arbitrageur who shorts technology companies and buys oil companies runs the risk that peace breaks out in the Middle East, causing the price of oil to plummet. In this case, perhaps the original judgment that oil stocks were cheap was correct but the bet loses money *ex post*.

In contrast, if A and B have identical cash flows but different prices, the arbitrageur eliminates fundamental risk. If securities A and B have other similar features, for example, similar liquidity, then bad-model risk is minimized as well. Violations of the law of one price are easier for economists to see and safer for arbitrageurs to correct. For example, suppose that A is a portfolio of stocks and B is a closed-end fund that owns A. If B has a lower price than A, then (when issues such as fund expenses are ignored) the arbitrageur can buy B, short A, and hope to make a profit if the prices converge. Unfortunately, this strategy is exposed to a third sort of risk, noise trader risk. An arbitrageur that buys the fund and shorts the underlying shares runs the risk that the discount may widen as investor sentiment shifts. This risk can be either systematic (all closed-end fund discounts widen) or idiosyncratic (Lee, Shleifer, and Thaler 1991). Since there is no guarantee that A and B will converge in price, the strategy is risky.

Noise trader risk can be eliminated in the long run in situations in which A and B are certain to converge in finite time. For example, suppose that at time T the closed-end fund B will liquidate, and all holders of B will receive a cash settlement equal to the net asset value of the portfolio, that is, A. We know that the prices of A and B will be identical at time T . Noise trader risk still exists in the intermediate period between now and T , but not over the long run. The terminal date eliminates other concerns as well; for example, liquidity is not an issue for investors holding until time T . In this case, with no fundamental risk, bad-model risk, or noise trader risk, there still is another problem that can cause the prices of A and B to be different: transactions costs (including trading costs and holding costs).

Both market efficiency and the law of one price are affected by transactions costs. If transactions costs are not zero, then arbitrageurs are prevented from forcing price all the way to fundamental value, and the same security can have different prices. In this case, then, Fama (1991, p. 1575) describes an efficient market as one in which "deviations from the extreme version of the efficiency hypothesis are within information and trading costs." An example is a market in which it is impossible to short a stock, equivalent to infinite transactions costs for short sales. In

this market, a stock could be massively overpriced, yet since there is no way for arbitrageurs to make money, the market is still efficient in the sense that there is no money left on the table. Still, this is market efficiency with very wrong prices.

In this paper we investigate apparent violations of the law of one price in which there are few risk issues involved, but transactions costs involved with short selling play an important role in limiting arbitrage. We study equity carve-outs in which the parent has stated its intention to spin off its remaining shares. A notable example is Palm and 3Com. Palm, which makes hand-held computers, was owned by 3Com, a profitable company selling computer network systems and services. On March 2, 2000, 3Com sold a fraction of its stake in Palm to the general public via an initial public offering (IPO) for Palm. In this transaction, called an equity carve-out, 3Com retained ownership of 95 percent of the shares. 3Com announced that, pending an expected approval by the Internal Revenue Service (IRS), it would eventually spin off its remaining shares of Palm to 3Com's shareholders before the end of the year. 3Com shareholders would receive about 1.5 shares of Palm for every share of 3Com that they owned.

This event put in play two ways in which an investor could buy Palm. The investor could buy (say) 150 shares of Palm directly or he could buy 100 shares of 3Com, thereby acquiring a claim to 150 shares of Palm plus a portion of 3Com's other assets. Since the price of 3Com's shares can never be less than zero (equity values are never negative), here the law of one price establishes a simple inequality: the price of 3Com must be at least 1.5 times the price of Palm. Since 3Com held more than \$10 a share in cash and securities in addition to its other profitable business assets, one might expect 3Com's price to be well above 1.5 times the price of Palm.

The day before the Palm IPO, 3Com closed at \$104.13 per share. After the first day of trading, Palm closed at \$95.06 a share, implying that the price of 3Com should have jumped to at least \$145 (with the precise ratio of 1.525). Instead, 3Com fell to \$81.81. The "stub value" of 3Com (the implied value of 3Com's non-Palm assets and businesses) was -\$63. In other words, the stock market was saying that the value of 3Com's non-Palm business was -\$22 billion! The "information costs" mentioned by Fama (1991) are small in this case, since the mispricing took place in a widely publicized IPO that attracted frenzied attention. The nature of the mispricing was so simple that even the dimmest of market participants and financial journalists were able to grasp it. On the day after the issue, the mispricing was widely discussed, including in two articles in the *Wall Street Journal* and one in the *New York Times*, yet the mispricing persisted for months.

This is a gross violation of the law of one price, and one for which

most of the risks identified above do not apply. An arbitrageur who buys 100 shares of 3Com and shorts 150 shares of Palm is essentially buying the 3Com stub for $-\$63$. If things go as planned, in less than a year this value must be at least zero. We do not need to agree on a model of asset pricing to agree on the proposition that one share of 3Com should be worth at least 1.5 shares of Palm. Noise trader risk is minimized because there is a terminal date at which the shares will be distributed. When the distribution occurs, the 3Com stub cannot have a negative price. Fundamental risks about the value of Palm are completely hedged. The only remaining problem is costly arbitrage. Still, investors were willing to pay over $\$2.5$ billion (based on the number of Palm shares issued) to buy expensive shares of Palm rather than buy the cheap Palm shares embedded in 3Com and get 3Com thrown in.

We do not claim that this mispricing creates exploitable arbitrage opportunities. To the contrary, we document the precise market friction that allows prices to be wrong, namely shorting costs. These costs arise when short sales are either very expensive or simply impossible. Although shorting costs are necessary in order for mispricing to occur, they are of course not sufficient. Shorting costs can explain why a rational arbitrageur fails to short the overpriced security, but not why anyone buys the overpriced security. To explain that, one needs investors who are (in our specific case) irrational, woefully uninformed, endowed with strange preferences, or for some other reason willing to hold overpriced assets. We shall refer to these conditions collectively as “irrational,” but they could be anything that causes a downward-sloping demand curve for specific stocks (despite the presence of cheaper and nearly identical substitutes).¹ Thus two things, trading costs and irrational investors, are necessary for mispricing. Trading costs, by limiting arbitrage, create an environment in which simple supply and demand intuition is useful in explaining asset prices. In our case, the demand for certain shares by irrational investors was too large relative to ability of the market to supply these shares via short sales, creating a price that was too high.

We investigate this question using all the cases we could find that share the key elements of the Palm-3Com situation, namely a carve-out with an announced intention to spin off the new issue in the near future. By limiting ourselves to these cases (as opposed to the much larger

¹ We use the term “irrational” for lack of a better word, but without wishing to engage in any deep philosophical debate about rationality. If someone buys a stock or bets on a horse because he or she likes the name and, in so doing, forgoes some financial benefit, we shall call that irrational, regardless of whether the utility derived from owning the asset with this name is large enough to compensate for the forgone financial advantage of owning a close substitute with a less desirable name. Since our paper concerns financial markets, it seems reasonable to equate nonpecuniary with irrational.

category of all carve-outs), we are able to minimize the risks that the spin-off never takes place and thus reduce the risk inherent in the arbitrage trade.

We start in Section II by describing carve-outs and spin-offs, showing how we construct the sample and describing its main features. In Section III we document high apparent returns that are implicit in market prices, describe relevant risks, and ask whether the high returns can plausibly be explained by risk. In Section IV we describe the short-sale constraints that allow mispricing to persist. We document another notable departure from the law of one price, the violation of put-call parity, and explain how this departure is consistent with short-sale constraints. In Section V we ask why stubs become negative, look at IPO day returns on parents and issues, and show the characteristics of investors in parents and issues.

II. Sample of Carve-outs

We examine carve-outs followed by spin-offs. An equity carve-out, also known as a partial public offering, is defined as an IPO for shares (typically a minority stake) in a subsidiary company. In an equity carve-out, a subsidiary firm raises money by selling shares to the public and then typically giving some or all of the proceeds to its parent. A spin-off occurs when the parent firm gives remaining shares in the subsidiary to the parent's shareholders; no money changes hands.

We study a sample of equity carve-outs in which the parent firm explicitly states its intention to immediately spin off its remaining ownership in the subsidiary. We study this sample of firms since in this case negative stubs appear to present a trading opportunity with fairly clear timing. In contrast, Cornell and Liu (2001), Schill and Zhou (2001), and Mitchell, Pulvino, and Stafford (2002) look at negative stub situations generally, not necessarily involving an explicit intention to spin off. Our focus on cases with a terminal date allows us to ignore some issues they discuss such as agency costs (the possibility that the parent firm may waste the cash generated by the subsidiary).

Spin-offs can be tax-free both to the parent firm and to its shareholders. In order to be tax-free, spin-offs need to comply with Internal Revenue Code Section 355, which requires that the parent (prior to the spin-off) owns at least 80 percent of the subsidiary. Thus if a firm plans a carve-out followed by a tax-free spin-off, it is necessary to carve out less than 20 percent of the subsidiary.

There are several reasons why a firm might carve out before spinning off. First, the parent firm might want to raise capital for itself (Allen and McConnell 1998). Second, the parent might wish to raise capital for the subsidiary to use. Third, the parent might want to establish a

dispersed base of shareholders in the subsidiary for strategic reasons related to corporate control (Zingales 1995). Fourth, a standard explanation is that the parent might want to establish an orderly market for the new issue by selling a small piece first (Cornell 1998). According to this explanation, the parent avoids flooding the market with a large number of new shares in a full spin-off, and the IPO gives an incentive for investment banks to market and support the new issue. Raising capital via a carve-out of the subsidiary, rather than an equity issue for the parent stock, is especially attractive if the firm believes that the parent stock is underpriced or the subsidiary will be overpriced, as in Nanda (1991) and Slovin, Sushka, and Ferraro (1995).

A. *The Sample*

We start building our sample by obtaining from Securities Data Corporation a list of all carve-outs in which the parent retains at least 80 percent of the subsidiary. Its list contains 155 such carve-outs from April 1985 to May 2000. To this list we added one issue (PFSWeb) that appears to have been miscoded by Securities Data and four issues occurring after May 2000. Using the Securities and Exchange Commission's (SEC's) Edgar database, we then searched registration form S-1 for explicit statements by the parent firm that it intended to distribute promptly the remaining shares to the shareholders. We discarded all firms for which we were unable to find a definitive statement that the parents intended to distribute all their shares. A typical statement, from Palm's registration, is that "3Com currently plans to complete its divestiture of Palm approximately six months following this offering by distributing all of the shares of Palm common stock owned by 3Com to the holders of 3Com's common stock." The statements often mentioned IRS approval as a precondition of distribution; the specified time frame for the distribution was usually six to 12 months.

We searched registrations starting in 1995, although since Edgar's database was incomplete prior to May 6, 1996, we were unable to find all firms before then. As it happens, we find no firms in 1995 that satisfied our requirements, so the final sample contains 18 issues from April 1996 to August 2000. This sample, shown in table 1, consists of every carve-out of less than 20 percent of subsidiary shares in which the parent declared its intention to distribute the remaining shares.

B. *Constructing Stubs*

We define the stub value using the ratio of subsidiary shares to be given to parent shareholders at the distribution date. The ratio is the parent's holdings of the subsidiary divided by the outstanding number of shares

TABLE 1
SAMPLE OF CARVE-OUTS

Issue Date	Parent	Subsidiary	Distribution Date	Negative Stub?
4/3/96	AT&T	Lucent Technologies	09/30/96	No
8/21/96	Tridex	TransAct Technologies	03/31/97	Marginal
11/13/96	Santa Fe Energy Resources	Monterey Resources	07/25/97	No
3/6/97	Odetics	ATL Products	10/31/97	Marginal
8/12/98	Cincinnati Bell	Convergys	12/31/98	No
12/3/98	Creative Computers	UBID	06/07/99	Yes
2/4/99	General Motors	Delphi Automotive Systems	05/28/99	No
8/10/99	Viacom	Blockbuster	Canceled	No
11/17/99	Hewlett-Packard	Agilent Technologies	06/02/00	No
11/17/99	HNC Software	Retek	09/29/00	Yes
12/1/99	Daisytek	PFSWeb	07/06/00	Yes
12/15/99	Metamor Worldwide	Xpedior	Canceled	Yes
3/1/00	3Com	Palm	07/27/00	Yes
4/3/00	Cabot Corp.	Cabot Microelectronics	09/29/00	No
6/26/00	Methode Electronics	Stratos Lightwave	4/28/01	Yes
6/26/00	Deluxe	Efunds	12/11/00	No
7/10/00	Eaton	Axcelis Technologies	12/29/00	No
8/9/00	Sea Containers	Orient Express Hotels		Marginal

NOTE.—List of 18 equity carve-outs, 1995–2000, in which the parent stated its intention to distribute to its shareholders its remaining shares in the subsidiary. Issue date is the pricing date for the IPO, occurring one day prior to the first day of trade. Distribution date is the date on which the spin-off is completed, occurring some time after the record date for the distribution.

of the parent on the record date of the distribution. Unfortunately, this ratio is not known with certainty on the issue date because the number of parent shares outstanding can fluctuate, for example, because of the conversion of convertible debt or the exercise of options owned by insiders.

Let the parent stock have a date 0 price per share of P_0^P and the subsidiary stock P_0^S . Let x be the ratio of subsidiary shares that are given to parent shareholders at the distribution date. A negative stub means that $S_0 = P_0^P - xP_0^S < 0$. We can also express the stub as a fraction of the parent, which we do with a lowercase s :

$$s_0 = \frac{P_0^P - xP_0^S}{P_0^P} = \frac{S_0}{P_0^P}.$$

Thus to calculate stub values, we have to estimate the expected ratio at each point in time. We did this in two stages. First, we simply used the naive ratio of the parent holdings in the subsidiary divided by the

current parent shares outstanding, using Center for Research in Security Prices (CRSP) data on shares outstanding. Since the various contingencies generally raise the number of shares of the parent, this naive ratio likely overstates the actual ratio and thus makes the calculated stub more negative. Second, after examining the pattern of stubs in the 18 cases, we more carefully studied the cases of potential negative stubs.

We concentrate on negative stubs in order to consider only cases of clear violations of the law of one price. Of course, there may be mispricing in other situations, but in such cases there is no uncontroversial proof of mispricing. So negative stubs should be considered the extreme cases of unambiguous mispricing. For the potential negative stubs, we gathered information that was available in real time to construct the estimated ratio. In all but one case the uncertainty about the final ratio appears to be small.²

Of our 18 firms, nine clearly had positive stubs. We classify three stubs as marginally negative. These were cases in which we observed small negative stubs on one or two days only or the correct ratio is sometimes unclear because of changing numbers of shares. For these cases we think that a reasonable person would not be convinced that the stub was negative given all available information. None of the three marginal cases involves a negative stub at or near the IPO date.

We identify six cases of unambiguously negative stubs: UBID, Retek, PFSWeb, Xpedior, Palm, and Stratos Lightwave.³ All six are technology stocks. UBID is an on-line auction firm. Retek produces business-to-business inventory software. PFSWeb provides transactions management services for e-commerce. Xpedior is an e-business consulting firm. Stratos Lightwave is an optical networking firm. Both the six parents and the six subsidiaries trade on NASDAQ.

As shown in table 2, for the six cases with negative stubs, four were negative at closing prices on the first day of trading, and the other two were negative by two days after. For five of the cases, the stub was negative for at least two months, with a maximum of 187 trading days for Stratos. For one case, Xpedior, the stub was negative for only two days before turning positive again. Xpedior's minimum stub also had a fairly small magnitude of only -19 percent of the parent company's value, unlike the other five, which had minimum stubs of -39 to -137 percent of

² In one case, Retek, there appears to have been substantial uncertainty about the final ratio since the parent's number of shares was somewhat volatile. Retek's parent ultimately decided to accelerate the vesting of the options held by insiders.

³ In these six cases, we calculated the estimated distribution ratio (prior to the actual distribution ratio announcement) as follows. For UBID and Retek, we used the ratio from the CRSP shares outstanding. For Palm and Stratos, we always used the ratios provided by Spin-off Advisors. For PFSWeb, we used the ratio provided by Spin-off Advisors until March 2000 and then used the CRSP shares. For Xpedior, we used the ratio provided by the company web page (in real time).

TABLE 2
STUB VALUES AND MARKET VALUES BETWEEN IPO AND DISTRIBUTION

	MIN STUB, S_p \$ PER PARENT SHARE	MIN STUB, s_p FRACTION OF PARENT	MAX MARKET VALUE OF ISSUE (\$ Millions)	TRADING DAYS			
				First Negative	Next First Positive	Last Negative	Distribution Announcement
Creative/UBID	-74.81	-1.37	342	1	114	113	113
HNC/Retek	-49.01	-.56	594	2	50	178	181
Daisytek/PFSWeb	-13.72	-.63	157	1	82	81	131
Metamor/Xpedior	-5.26	-.19	315	3	5	92	67
3Com/Palm	-63.16	-.77	2,514	1	48	47	47
Methode/Stratos	-20.95	-.39	499	1	133	146	187

NOTE.—The stub value in dollars per share is $S_i = P_i^p - x_i P_i^s$. The stub value as a fraction of parent value is $s_i = (P_i^p - x_i P_i^s) / P_i^p = S_i / P_i^p$. The first day of trading is day 1. Min stub is that between IPO and distribution. Max market value of issue is the maximum price times the number of shares issued (not outstanding) during the interval between first negative and next first positive. First negative is the first trading day with a negative stub. Next first positive is the subsequent day on which prices imply a positive stub. Last negative is the last day on which a negative stub occurs. All calculations are based on closing prices. For Metamor/Xpedior, day 67 is the day on which the takeover of parent Metamor is announced.

the parent's value. Thus Xpedior is a much weaker case in terms of the persistence and magnitude of the mispricing.

Table 2 shows the magnitude of the mispricing in a variety of ways. Perhaps the most relevant is the market value of the shares trading in the subsidiary. This number (which uses the number of publicly trading shares, not the number of outstanding shares) is at its peak, \$2.5 billion, for Palm, meaning that investors worth \$2.5 billion thought it was better to own Palm than to own 3Com.

C. *Time Pattern of Negative Stubs*

Figures 1–4 show the time series of stub values for the six cases of negative stubs. Except in figure 2, the solid line shows the stub prior to distribution and the dashed line shows the parent share price after the distribution. Several patterns are apparent. First, stubs start negative and gradually get closer to zero, eventually becoming positive. Second, the announcement of IRS approval and the consequent announcement of a distribution date (occurring on the same day) cause the stub to go from negative to positive in two cases, UBID and Palm. Thus in these cases the market is acting as though there is significant news on these days.⁴

In one case, Xpedior, the distribution never occurred. On March 22, 2000, the parent company Metamor announced that another firm was acquiring it. Xpedior's stub rises markedly on this day. However, one could argue that on and after this date, Xpedior's stub has little meaning since the distribution is presumably canceled. On the announcement day, although not explicitly canceling the spin-off, the acquirer failed to confirm the spin-off and instead announced that it had gained control of Xpedior and was investing additional money in it.

The picture is one of predictable idiosyncratic movement in stubs. Stubs start off negative and then get positive. This pattern is repeated over time and does not appear to reflect systematic exposure to some common factor, but rather idiosyncratic developments.

We draw two conclusions from the analysis so far. First, we are able to identify six cases of clearly negative stubs. We do not think that the proportion of negative stubs, one-third of the cases we study, is particularly significant. As we stressed above, a negative stub indicates a gross case of mispricing. Even a single case would raise important questions

⁴ In one case, Retek, the stub has less of a trend. Because of the uncertainty about the ultimate distribution ratio, Retek's true stub is not totally clear in July 2000. The reaction of the stub to the distribution announcement has a different meaning for Retek since the announcement contains important quantitative information. The announced ratio was 1.24, whereas 10 days earlier an analyst report contains an estimate of 1.40.

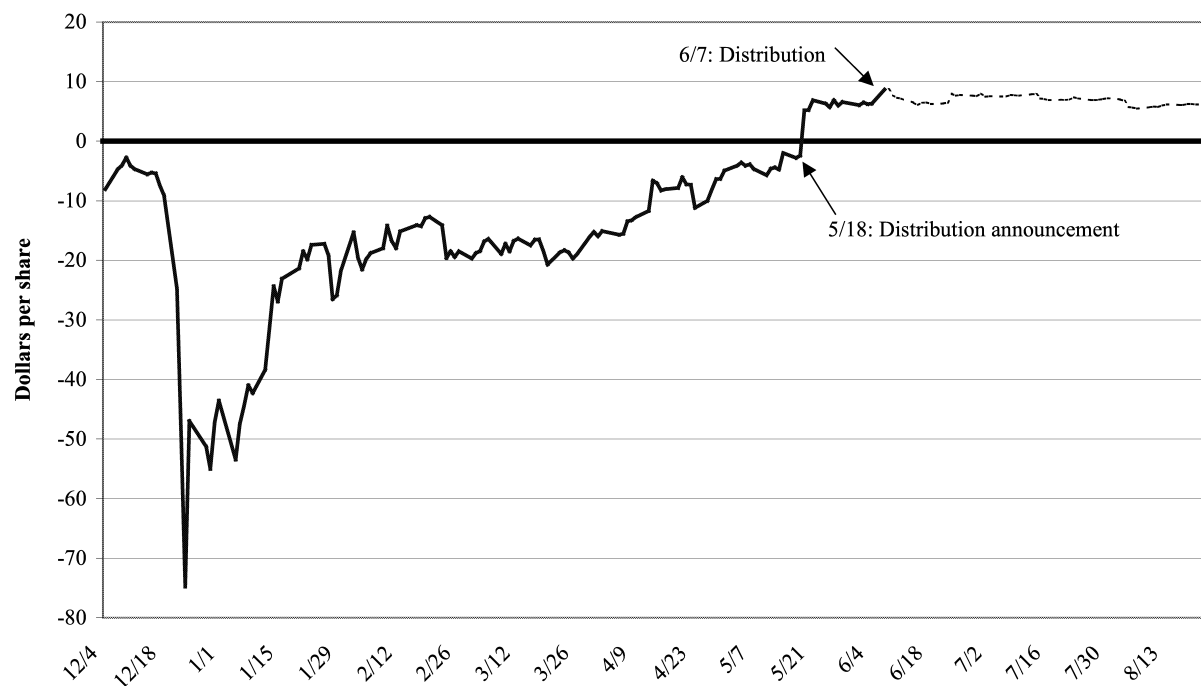


FIG. 1.—Creative Computers/UBID stub, December 4, 1998–August 24, 1999

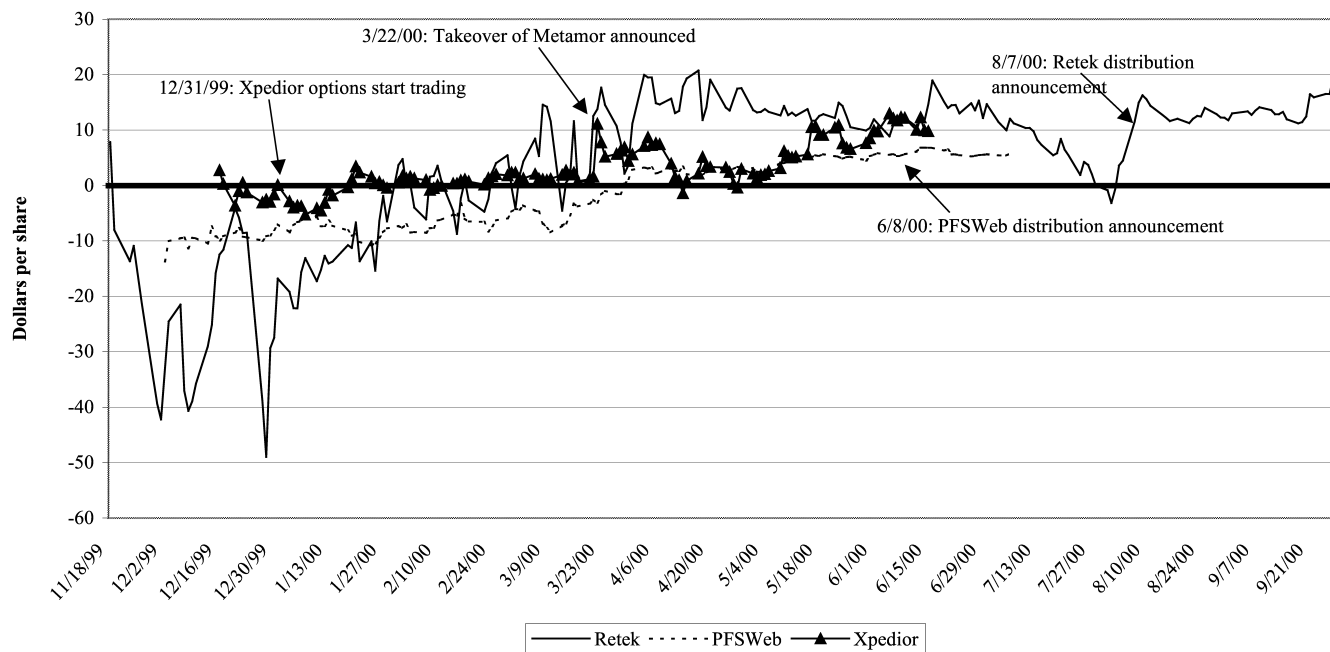


FIG. 2.—Stub values for HNC/Retek, Daiseytek/PFSWeb, and Metamor/Xpedior

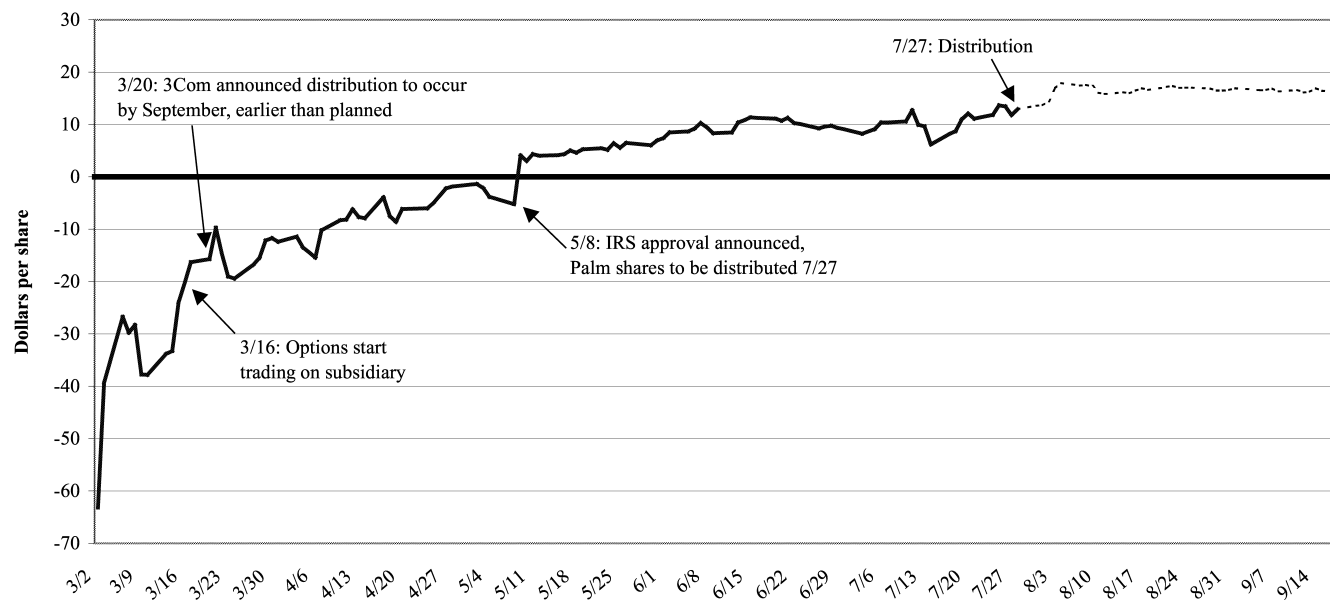


FIG. 3.—3Com/Palm stub, March 2, 2000–September 18, 2000

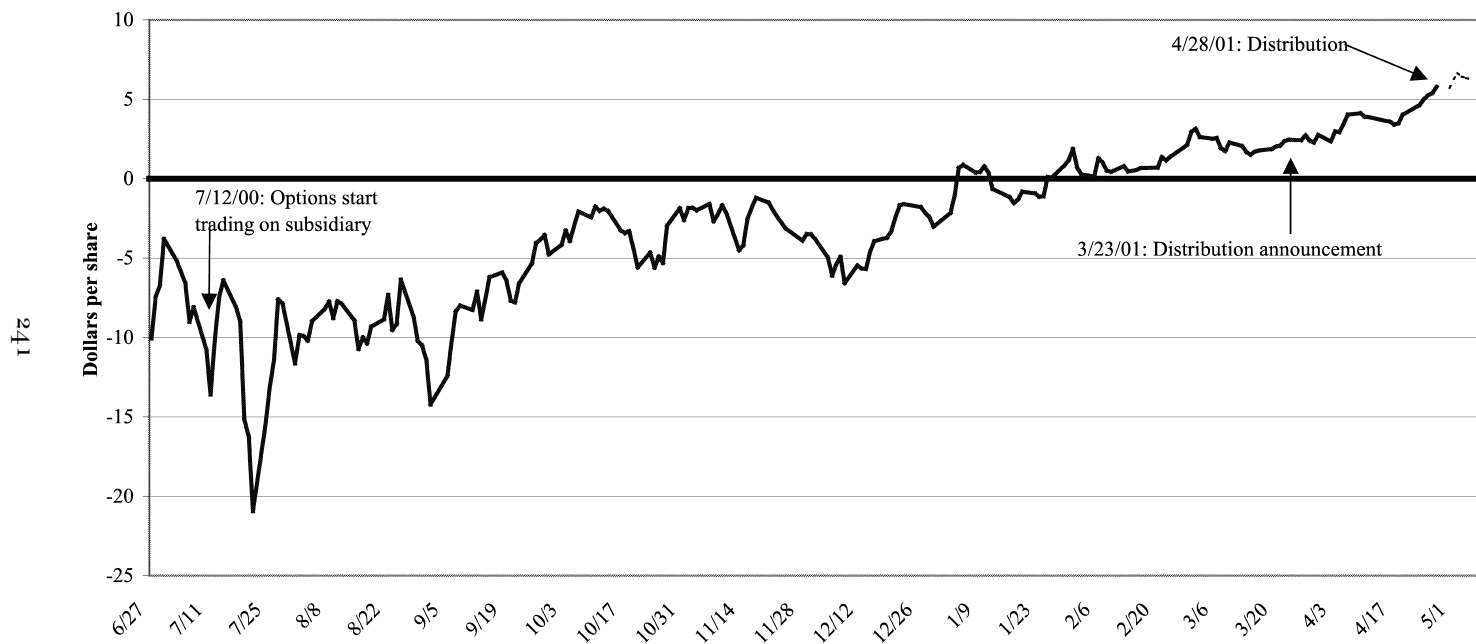


FIG. 4.—Methode/Stratos stub, June 27, 2000–May 7, 2001

about market efficiency. The fact that we find six such cases indicates that the highly publicized Palm example was not unique.⁵

Second, all the cases we study show a similar time pattern of returns whereby the stub becomes less negative over time and eventually becomes positive. This suggests that market forces act to mitigate the mispricing, but slowly. We return to this slow adjustment, which reflects the difficulty of shorting due to the sluggish functioning of the market for lending stocks, in Section IV.

III. Risk and Return on Stubs

In this section, we investigate the returns to an investment strategy of buying the parent and shorting the subsidiary. We find that this strategy produces high returns with low (and largely idiosyncratic) risk. However, we caution readers not to rush out to form hedge funds to exploit this phenomenon; as we show in the next section, the high returns we find on paper are probably not achievable in practice because of the difficulty of shorting the subsidiary (although we are aware of individual investors who did make money on these situations). Thus the question we ask is whether the investment strategy would have produced profits if it could have been implemented.

This investment strategy is related to several controversies in finance: value, IPOs, and the diversification discount. First, it is a value strategy of buying cheap stocks and shorting expensive stocks. Second, it is a strategy that shorts IPOs. Ritter (1991) documents that IPOs tend to have low subsequent returns, but the statistical soundness of this finding has been the subject of a vigorous debate summarized in Fama (1998) and Loughran and Ritter (2000). In a subset of the IPO debate, Vijh (1999) finds that carve-out stocks do not have low subsequent returns. Third, it is a strategy that buys firms with a large diversification discount. Lamont and Polk (2001) show that the diversification discount partially reflects subsequent returns on diversified firms, so that the diversification discount does not reflect only agency concerns such as wasteful managers. In the case of our firms, it seems unambiguous that mis-

⁵ It is hard to say whether the ratio of one-third overestimates or underestimates the prevalence of mispricing. On the one hand, perhaps firms tend to do carve-outs when they think that their subsidiaries are overpriced, in which case the 18 firms are not a representative sample (firms should issue equity when that equity is overpriced, as argued by Stein [1996]). Further, it could be that 1998–2000 was a time in which mispricing was prevalent, but in most years mispricing is rare. Ritter and Welch (2002) show that this period was one with extraordinary IPO first-day returns, and Ofek and Richardson (2001) show that Internet-related IPOs had especially high first-day returns in this period. On the other hand, mispricing could occur more than one-third of the time. We show that only six of the 18 have negative stubs. Perhaps the other 12 have stubs that are too low or too high. So in that sense, perhaps one-third is a lower bound for relative mispricing.

pricing drives the subsequent pattern of returns, so that we have a clear example in which the value/IPO/diversification effect is due to mispricing.

A. *Returns on Stub Positions*

The following analysis ignores dividends and assumes that the distribution takes place with a fixed distribution ratio at time T . First, since the stub must go from negative to positive by date T , it must be the case that $R_T^P > R_T^S$, where R_T^P and R_T^S are the returns on the parent and subsidiary between date 0 and date T . Thus if an investor buys the parent and shorts an equal dollar amount of the subsidiary, she gets a positive return of $R_T^P - R_T^S$. In a frictionless market in which the investor gets access to short-sale proceeds, this strategy is a zero-cost or self-financing strategy. For this strategy, the exact distribution ratio x is not important, as long as one knows that the stub is negative initially. On paper, this strategy is an arbitrage opportunity, since it has zero cost and generates strictly positive cash flow in the future.

Assuming that the distribution takes place with known ratio x , one can construct a position that is a pure bet on the stub. This second strategy eliminates the effect of fluctuations in subsidiary value and again guarantees strictly positive returns. It buys one share of the parent, shorts x shares on the subsidiary, and (again with access to the short-sale proceeds ignored) invests the resulting $-S_0$ dollars of cash in the initial period at the risk-free rate of R_f . Again, this strategy is theoretically self-financing and puts equal amounts into the long portfolio (consisting of riskless assets and the parent) and the short portfolio (consisting of the subsidiary). One can express the returns on this strategy as

$$\frac{1}{1 - s_0} R_T^P + \frac{-s_0}{1 - s_0} R_T^F - R_T^S.$$

Table 3 shows returns from the strategy of buying the parent and shorting the subsidiary at the closing price on the first day on which the stub is negative. We examine two holding periods: holding until one day after the announcement date and holding until one day before the distribution date. For the purposes of table 3, we use the takeover announcement for Xpedior's announcement date and the takeover consummation as Xpedior's distribution date. Table 3 shows that parents had returns that were 30 percent higher than subsidiaries holding until the announcement date and 33 percent higher holding until the distribution date. This difference was statistically significant. From this evidence alone, one cannot say whether the subsidiary is overvalued or

TABLE 3
TOTAL RETURNS FROM FIRST NEGATIVE STUB TO ANNOUNCEMENT/DISTRIBUTION DAYS

	FIRST NEGATIVE STUB	ANNOUNCEMENT DAY PLUS ONE				DISTRIBUTION DAY MINUS ONE			
		Stub	R_t^p	R_t^s	$R_t^s - R_t^p$	Stub	R_t^p	R_t^s	$R_t^s - R_t^p$
Creative/UBID	-8.09	5.17	.49	.00	.49	6.32	.25	-.22	.47
HNC/Retek	-8.04	14.94	-.19	-.34	.15	17.80	.23	.09	.13
Daisytek/PFSWeb	-13.72	5.64	-.48	-.84	.36	5.39	-.59	-.90	.31
Metamor/Xpedior	-3.65	7.75	.15	-.22	.37	9.89	-.06	-.48	.41
3Com/Palm	-63.16	4.09	-.41	-.69	.28	13.43	-.17	-.61	.44
Methode/Stratos	-10.06	2.42	-.59	-.72	.13	5.79	-.59	-.78	.20
Average	-17.78	6.67	-.17	-.47	.30	9.77	-.16	-.48	.33
	(-1.94)	(3.70)	(-1.02)	(-3.46)	(5.21)	(4.79)	(-1.03)	(-3.20)	(5.78)
Average excluding Xpedior	-20.61	6.45	-.24	-.52	.28	9.75	-.18	-.48	.31
	(-1.93)	(2.94)	(-1.23)	(-3.36)	(4.18)	(3.90)	(-.95)	(-2.62)	(4.69)

NOTE.—Returns are total simple returns from the day of the first negative stub to either the day after the announcement day or the day prior to the distribution day. For Xpedior, we count the announcement day as the day on which Xpedior's parent announces that it is being acquired, and the distribution day as the day Xpedior's parent ceases trading. *t*-statistics of averages are in parentheses.

TABLE 4
CAPM AND THREE-FACTOR REGRESSION FOR MONTHLY TRADING
STRATEGIES

	SIMPLE STRATEGY		HEDGED STRATEGY	
	(1)	(2)	(3)	(4)
α	.10 (.03)	.10 (.03)	.09 (.03)	.09 (.03)
RMRF	1.22 (.53)	1.41 (.60)	.89 (.47)	1.06 (.53)
HML		.46 (.45)		.42 (.40)
SMB		.47 (.63)		.43 (.56)
R^2	.22	.27	.16	.21

NOTE.—Monthly regressions of strategy returns on factors. Calculations use closing prices. The strategy takes a position on the last day of the month if the stub is negative on that day and holds until the last day of the month prior to the distribution month. In all five cases, the position is initiated at the end of the first month of trading. Since Metamor/Xpedior does not have a negative stub at the end of the month, it is not included in this strategy. Equal-weighted returns are on from one to three paired positions per month. The simple strategy is $R_t^p - R_t^s$. The hedged strategy is

$$\frac{1}{1-s_0} R_t^p + \frac{-s_0}{1-s_0} R_t^e - R_t^s$$

where R_t^p is the monthly return from the parent stock and R_t^s is the monthly return from the subsidiary stock; $s_0 = (P_0^e - xP_0^p)/P_0^p$ is the stub value as a percentage of the parent stock price, as of the last day of the first month of trading. RMRF is the CRSP value-weighted market return minus Ibbotson Treasury bill return. HML and SMB are the value and size factors from Fama and French (1993) and come from the web page of Kenneth French. HML is the returns on stocks with high book to market ratios minus the returns on stocks with low book to market ratios. SMB is the return on small-cap stocks minus the returns on big-cap stocks. The number of observations is 21 months. Standard errors are in parentheses.

the parent is undervalued. Later, we show evidence from options markets implying that it is the subsidiary that is overpriced.

B. Traditional Measures of Risk

Table 4 uses monthly calendar time portfolio returns reflecting a strategy of buying parents and shorting subsidiaries. On the last trading day of the month, if the subsidiary has a negative stub on that day, we buy the parent and short the subsidiary. We maintain this position until the last day of the month prior to the distribution date. We calculate equal-weighted returns on the portfolio holdings on this strategy. The strategy holds one to three paired positions each month, for the 21 months of returns from January 1999 to May 1999 (UBID) and December 1999 to March 2001 (the other four subsidiaries; the strategy does not take a position in Xpedior).

Over this period, the simple strategy of buying parents and shorting subsidiaries in equal dollar amounts has a monthly return that averages 10 percent per *month* (significantly different from zero) with a standard

deviation of 14 percent per month, producing a monthly Sharpe ratio of 0.67 per month. The hedged strategy that takes a pure bet on the stub has a slightly higher Sharpe ratio of 0.70 a month. Over the same period, the average market excess return (value-weighted New York Stock Exchange/American Stock Exchange/NASDAQ return from CRSP minus Treasury bill returns from Ibbotson Associates) was negative. From July 1927 to March 2001 the market had a Sharpe ratio of 0.12. Thus stub strategies have risk-return trade-offs more than four times more favorable than the market's.

Table 4 shows estimates of a capital asset pricing model (CAPM) equation. Although the strategy has a positive and significant market beta (so that subsidiaries have more market risk than parents), α is a huge 10 percent per month for the simple strategy and 9 percent for the hedged strategy. The t -statistic on α formally tests the hypothesis that the stub trading strategy can be used to produce a higher Sharpe ratio than the market. Even using these highly undiversified portfolios with only 21 monthly observations, we are able to resoundingly reject the hypothesis that α is zero. Using the three-factor model of Fama and French (1993) does not change the conclusion.

C. *Risks Specific to Stubs*

Since our sample is so small, it is useful to discuss some events that did not occur but might be expected to occur in a larger sample. Events that might have a negative impact on arbitrage investors include canceling the spin-off or changing the distribution ratio by lowering the number of subsidiary shares that each parent shareholder receives. If the expected ratio changes, then the stub can go from negative to positive without any change in prices.

Cancellation of the distribution can occur for several reasons. First, if the firm does not receive IRS approval, the spin-off is not tax-free and will probably be canceled. Our impression is that IRS rejection is a low-probability event. Second, the firm might change its mind and cancel the spin-off even if the IRS does approve. Although the parents in our sample stated their intention to distribute their ownership, this statement is not legally binding. An example that occurred in our larger sample of 18 carve-outs is Blockbuster. The parent, Viacom, stated in an SEC filing four months after the carve-out that it would wait until Blockbuster's share price was higher before completing the separation. In this example, Viacom's decision is not much of a negative event for the stub strategy of shorting the subsidiary, since the distribution is canceled only in the state of the world in which the subsidiary price remains low. Nevertheless, it is always possible for a canceled spin-off to cause the trading strategy to reap negative returns.

Another reason a distribution can be canceled is a takeover by a third party or shareholder pressure. We have already discussed the case of Xpedior, whose parent was acquired. As shown in table 3, this acquisition did not prevent the stub strategy from earning high returns. Another example from our sample is PFSWeb. Prior to the carve-out the parent firm received an unsolicited takeover bid that was conditional on canceling the spin-off, and later a large shareholder in the parent publicly objected to the spin-off and threatened legal action. Despite these events, the carve-out and distribution took place as planned.

These examples highlight the fact that the trading strategy is not riskless. It is worth noting, however, that many of these unpredictable events seem likely to benefit strategies that buy parent shares and short subsidiary shares. Takeover of the parent company (with the usual takeover premia), shareholder pressure to increase value to parent shareholders, or cancelation of the distribution due to low prices of the subsidiary all are positive for the strategy.

Since returns are high and the risks seem both quite low and almost entirely idiosyncratic, it appears that these subsidiaries are overpriced relative to the parent shares. However, with only six pairs of firms and only 21 months of returns, this evidence is not conclusive. It is possible that there was some negative event capable of generating large losses to the arbitrage strategy that just did not come up during the period we studied. To address these concerns we shall turn to the options market for additional evidence on mispricing.

IV. Short-Sale Constraints and the Persistence of Mispricing

The previous section argued that the negative stub situations created very attractive investment opportunities. Why, then, didn't rational arbitrageurs step in to correct the mispricing by buying the parent and shorting the subsidiary? There are many types of reasons that in general might prevent rational investors from correcting mispricing. These reasons include fundamental risk, noise trader risk, liquidity risk, institutional or regulatory restrictions, and tax concerns. Shleifer and Vishny (1997) discuss idiosyncratic risk and agency problems in delegated portfolio management (see also Pontiff 1996). In the cases we study, the principal idiosyncratic risk is the possibility that the distribution will not take place, and consistent with this idea, when the distribution date is announced, the stub values sometimes go from negative to positive. This pattern is consistent with arbitrageurs who are reluctant to take on substantial idiosyncratic risk.

In many situations, noise trader risk, institutional restrictions, and so forth might cause assets to be mispriced. In our specific case, however, these issues appear to be minimal, and the chief impediment to arbitrage

is short-sale constraints. First, shorting can be simply impossible. Second, when shorting is possible, it can have large costs.

A. *Description of the Shorting Process*

The market for shorting stock is not simply the mirror image of buying stocks long, for various legal and institutional reasons. To be able to sell a stock short, one must borrow it; and because the market for borrowing shares is not a centralized market, borrowing can be difficult or impossible for many equities. In order to borrow shares, an investor needs to find an institution or individual willing to lend shares. Financial institutions, such as mutual funds, trusts, or asset managers, typically do much of this lending. These lenders receive a fee in the form of interest payments generated by the short-sale proceeds, minus any interest rebate that the lenders return to the borrowers. Stocks that are held primarily by retail investors, stocks with low market capitalization, and illiquid stocks can be more difficult to short.

Being simply unable to short is particularly likely for individual retail investors, although there is extensive anecdotal evidence of institutional investors unable to short the overpriced subsidiaries. Regulations and procedures administered by the SEC, the Federal Reserve, the various stock exchanges, and individual brokerage firms can mechanically impede short selling, especially immediately after the IPO. In some cases, firms ask their stockholders not to lend their stock, to prevent short sellers from driving down the price. In the specific case of Palm, the *Wall Street Journal* reported that “it may be possible to short sometime next week The brokerage firms and institutional investors that control much of Palm’s stock generally agree not to immediately lend the stock to short sellers until sometime after the IPO date” (March 6, 2000, p. C15).

For institutions that are able to find shares to borrow, the cost of shorting is reflected in the interest rate rebate they receive on the short-sale proceeds. This rebate acts as a price that equilibrates supply and demand in the securities lending market. The rebate can be negative, meaning that institutions that sell short have to make a daily payment to the lender for the right to borrow the stock (instead of receiving a daily payment from the lender as interest payments on the short-sale proceeds). This rebate apparently only partially equilibrates supply and demand, because the securities lending market is not a centralized market with a “market-clearing” price. Instead, rebates reflect individual deals struck among security owners and those wishing to short, and these actors must find each other. This search may be costly and time-consuming (Duffie [1996] suggests that the securities lending market could be described by a search model).

B. Shorting Costs and Overpricing

Short-sale constraints have long been recognized as crucial to the workings of efficient markets. Diamond and Verrecchia (1987) describe a model with some informed traders, other uninformed but rational traders, and possible restrictions on shorting. In their model, although short-sale constraints impede the transmission of private information, short-sale constraints do not cause any stocks to be overpriced. Uninformed agents rationally take into account short-sale constraints and set prices realizing that negative opinion may not be reflected in trading.

With irrational traders, however, short-sale constraints can cause some stocks to become overpriced. With short-sale constraints, rational arbitrageurs can refrain only from buying overpriced stocks, and if there are enough irrational traders, stocks can be overpriced (see, e.g., Miller 1977; Russell and Thaler 1985; and Chen, Hong, and Stein 2002). A variety of evidence is consistent with such overpricing. Figlewski and Webb (1993) and Dechow et al. (2001) show that stocks with high short interest have low subsequent returns. Jones and Lamont (2002) show that stocks that are expensive to short or enter the lending market have high valuations and low subsequent returns.

Miller (1977) describes how short-sale constraints can cause prices to reflect only the views of optimistic investors. In describing the types of stocks likely to be overpriced because of divergence of opinion, he presciently lists many of the characteristics of our sample: IPOs with short operating history and exciting new products. He discusses how short-sale constraints might explain the diversification discount; our firms are extreme examples of such discounts.

One potentially confusing aspect of short sales is that the cost for those borrowing the stock is income for those lending the stock. Thus it is not quite accurate to say that only an irrational investor would buy an overpriced stock. A rational investor might be willing to buy an overpriced stock if he can derive sufficient income from lending it to short sellers. On the basis of this fact, one might be tempted to conclude that the situation we observe is therefore “rational,” since rational investors are willing to buy the subsidiaries. Along these lines, one could argue that the observed returns for Palm, for example, are not a “real” return since the true return should include the income from lending (reflecting the convenience yield or dividend from securities lending) and that the “marginal” investor sets the traded price to embody all income generated by the shares.

Such an interpretation would be a mistake. It is important to recognize that irrationality, or at least some nonstandard phenomena causing downward-sloping demand curves for stocks, is a crucial element to any explanation of the facts we are studying. Consider the following ex-

ample. A firm, consisting of \$100 in cash, issues 100 shares. The firm will liquidate tomorrow, and each share will pay a liquidating dividend of \$1.00. These shares are issued and sold by auction to investor *I*, who buys all 100 shares directly from the firm. Investor *I* mistakenly believes that the shares will pay out \$2.01 tomorrow and “wins” the auction with a bid of \$2.00 per share. It is clear in this example that investor *I* has overpaid for the shares and that \$2.00 is a “real price.”

Now suppose that two other investors, *Y* and *Z*, enter the market. Investor *Y* buys all 100 shares from the firm for \$2.00 and lends them to *Z*. Investor *Z* pays *Y* a fee of \$1.00 for each share lent and sells the shares to *I* for \$2.00. Now in this example, *Y* and *Z* are both acting rationally. However, there is no sense in which *Y* and *Z* are the “marginal” investors that set prices. Investors *Y* and *Z* would be just as happy with a price of \$200 per share (and a corresponding loan fee of \$199). It is the willingness of investor *I* to overpay that sets the price of the shares. The price of \$2.00 is a real price, and the firm should rationally respond to the mispricing by issuing more shares. The fact that *Y* and *Z* are intervening actors between the firm and the owner is irrelevant in this example.

The number of shares not lent out must equal the number of shares outstanding; it is always true that *someone* has to own the shares issued by the firm; not all owners can lend their shares. If the firm issues 100 shares, exactly 100 shares have to be owned by someone who is not lending them out. Thus it is not an empirical issue whether the owners of Palm lent out their shares or not, but rather a simple identity: \$2.5 billion worth of shares were owned by investors who were not receiving any lending income from their shares.

More generally, in any situation in which the shorting market is imperfect and some investors have a downward-sloping demand curve for a particular security, equilibrium prices depend on supply and demand. For example, Duffie (1996) and Krishnamurthy (2002) study the market for Treasury bonds. At some times, the price of on-the-run Treasury bonds is particularly high relative to off-the-run bonds, perhaps reflecting liquidity concerns. At these times, the cost of shorting reflects these price differences, so that it is not necessarily profitable to short the expensive bond and buy the cheap one, and it might well be rational to buy the expensive bond in order to reap the lending income. These price movements reflect the existence of a demand curve for on-the-run securities. In a frictionless market, arbitrageurs would be able to supply bonds to meet this demand for on-the-run securities. Similarly, in our example, if investor *Z* was able to manufacture new shares, he might be able to satiate investor *I*.

TABLE 5
PERCENTAGE SHORT INTEREST

	FIRST MONTH		SECOND MONTH:	PEAK:
	Parent	Subsidiary	Subsidiary	Subsidiary
Creative/UBID	4.2	8.5	54.7	70.9
HNC/Retek	7.5	19.8	37.4	53.4
Daisytek/PFSWeb	1.6	17.7	48.6	63.7
Metamor/Xpedior	4.9	17.2	24.6	26.8
3Com/Palm	2.6	19.4	44.9	147.6
Methode/Stratos	1.5	31.8	50.3	114.7
Average	3.7	19.1	43.4	79.5
Difference from previous column		15.3	24.3	36.1
t-statistic		4.4	4.5	2.3

NOTE.—Short interest calculated as a percentage of parent shares outstanding or subsidiary shares trading. The level of short interest comes from the National Association of Securities Dealers and occurs on or prior to the fifteenth calendar day of the month. The shares outstanding of the parent are taken from CRSP, and the shares issued in the IPO are taken from company SEC filings. First month is the first observed short interest after the IPO, and second month is one month later. Peak is the highest level between the IPO date and the distribution date.

C. Evidence on Short Sales

Given the obvious nature of the mispricing in the cases of negative stubs and the publicity associated with some of the cases such as Palm, it is not surprising that many investors were interested in selling the subsidiaries short. Table 5 shows the level of short interest for parents and subsidiaries. Short interest is much higher in subsidiaries than in parents, consistent with the idea that the subsidiaries are overpriced. For parents, we report short interest divided by total shares outstanding. For subsidiaries, we report short interest divided by total shares sold to the public in the IPO, since these shares are the only ones trading in the market.

Table 5 shows that on the first reporting date after the IPO, the parents had an average of 3.7 percent of their shares shorted. The subsidiaries had a significantly larger short interest of 19.1 percent. A month later, on the second reporting date, 43.4 percent of subsidiary shares were shorted. This dramatic increase over time could be produced by some combination of two factors. First, it may take a while for investors to become aware of the mispricing and decide to try to exploit it. Second, and more plausibly, the short-sale market works sluggishly. Only shares that are held by institutions willing to lend them are available for interested short sellers, and it takes time for lendable shares to find their way to the market for shorting.

Table 5 also shows the peak level of short interest for subsidiaries, for the time between the IPO and the distribution date. At the peak, short sales are 79.5 percent of total shares trading, and for Palm the level is an amazing 147.6 percent. More than all the floating shares had been

sold short. This is possible if shares are borrowed and then sold short to an investor who then permits the shares to be borrowed again. Again, the multiplier-type process takes time to operate because of frictions in the securities lending market. This peak level of short interest for Palm was reached on July 14, 2000, two weeks before the announced distribution, at a time when the stub was positive but rising.

Figures 5 and 6 show short interest (expressed as a percentage of total shares issued) and stub value (expressed in dollars per parent company stock price) for Palm and Stratos over the relevant period. The figures show that as the supply of shares available grows via short sales, the stub value gets more positive. One might interpret this pattern as roughly tracing out the demand curve for the overpriced subsidiary. As the supply of shares grows via short sales, we move down the demand curve of irrational investors and the subsidiary price falls relative to the parent.

Although quantity data in the shorting market are readily available, price data are not. We do not know precisely what the cost of shorting the overpriced subsidiaries was. We do have scattered evidence for four of the six subsidiaries. D'Avolio (2002) reports maximum borrowing costs of 50 percent (in annual terms) for Stratos Lightwave in December 2000, 35 percent for Palm in July 2000, and 10 percent each for PFSWeb in June 2000 and Retek in September 2000.⁶

Given these high lending fees, it might seem surprising that it took so long for owners to offer their shares to the lending market. This apparent money left on the table reflects the dysfunctional nature of the securities lending market. The system is just not set up to facilitate lending of shares held primarily in retail accounts. Further, most of the outstanding shares in the subsidiaries were held by parents, and there are several reasons why parents do not lend out their subsidiary shares. First, it is typically the case that the shares owned by parents are not registered prior to the actual distribution to the shareholders and therefore cannot be publicly traded. Second, lending shares might breach the fiduciary duties that parent directors owe to the parent or subsidiary. Third, lending shares could jeopardize the ability to do a spin-off tax-free by calling into question the "independent business purpose" of the spin-off, or by reducing the parent's control below important tax thresholds.

We next look at options markets to get more complete quantitative evidence on just how expensive it is to sell short.

⁶ With the exception of Stratos Lightwave (which has a distribution date occurring after D'Avolio's sample ends), all these dates are on or near the distribution date.

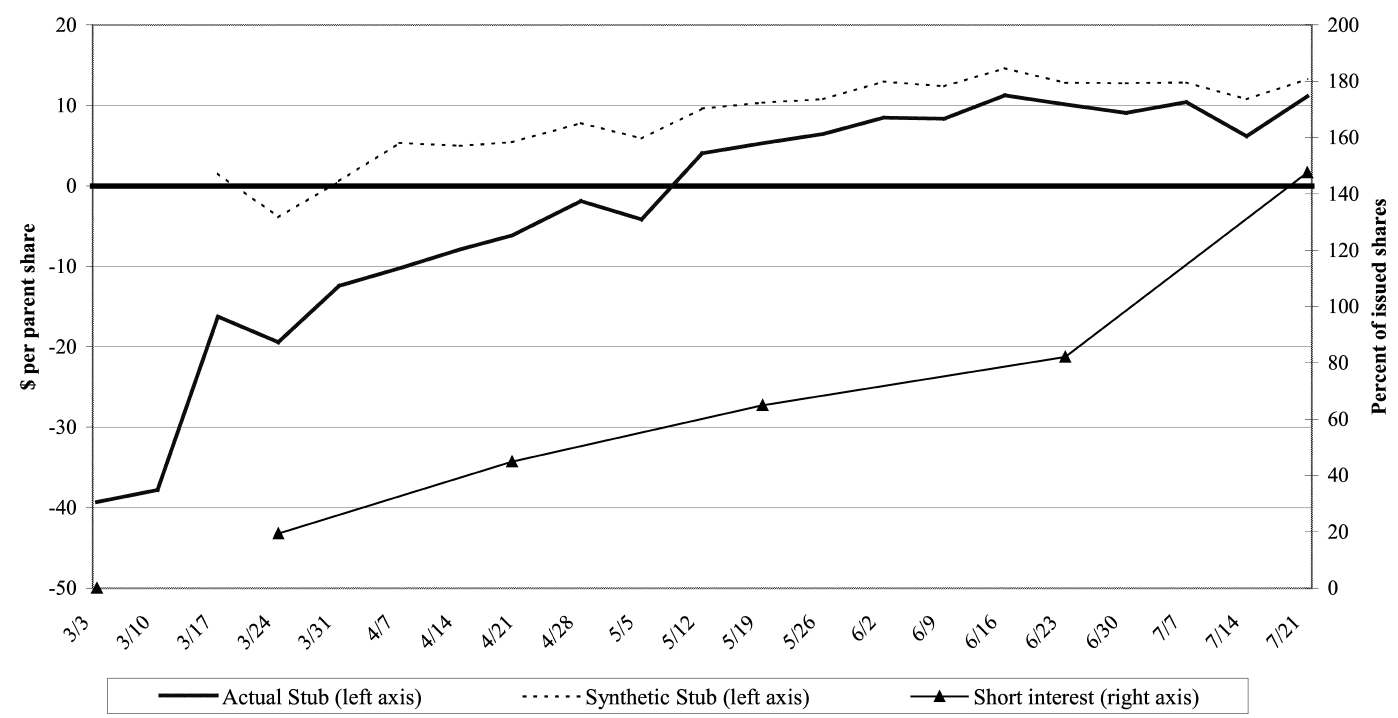


FIG. 5.—3Com/Palm: actual stub, synthetic stub, and short interest, March 3, 2000–July 21, 2000

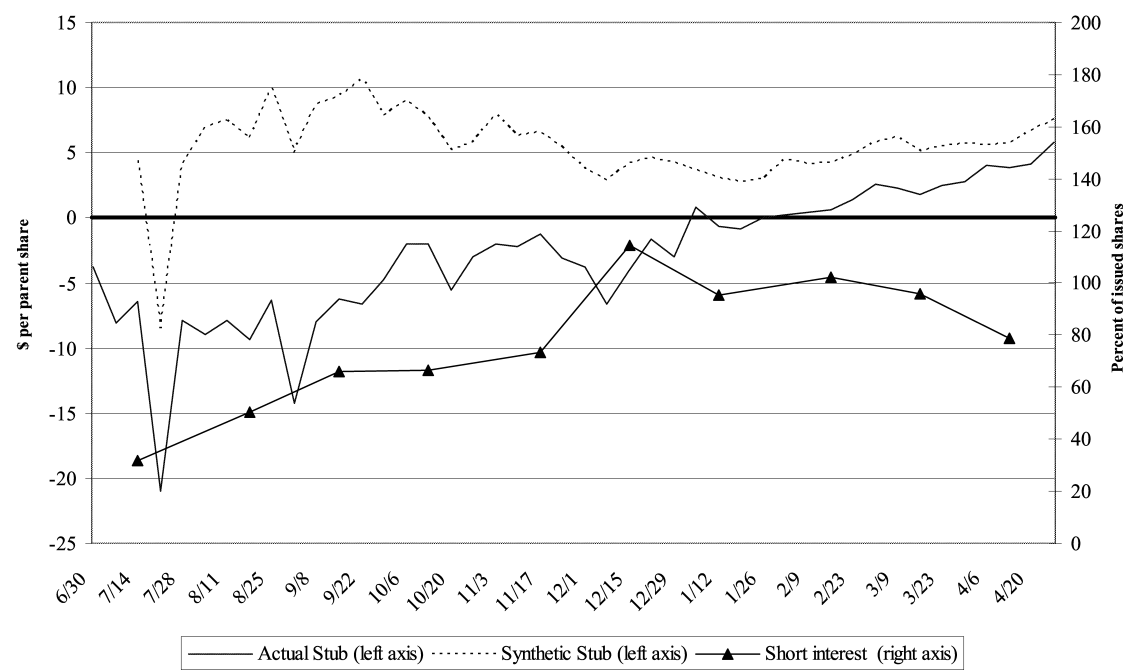


FIG. 6.—Methode/Stratos: actual stub, synthetic stub, and short interest, June 3, 2000–April 27, 2001

D. Short-Selling Constraints: Evidence from Options

Options can facilitate shorting, both because options can be a cheaper way of obtaining a short position and because options allow short-sale-constrained investors to trade with other investors who have better access to shorting. Figlewski and Webb (1993) show that optionable stocks have higher short interest. Sorescu (2000) finds that in the period 1981–95, the introduction of options for a specific stock caused its price to fall, consistent with the idea that options allow negative information to become impounded into the stock price.⁷

In a frictionless market, one expects to observe put-call parity. It should hold exactly (within trading costs) for European options and approximately for American options. One way of expressing put-call parity is to say that synthetic shares (constructed using options plus borrowing and lending) should have the same price as actual shares, plus or minus trading costs such as the bid/ask spread. This equality is just another application of the law of one price. A weaker condition than put-call parity, which should always hold for non-dividend-paying American options, is the following inequality: the call price minus the put price is greater than the stock price minus the exercise price. For options that are at-the-money (so that the option's exercise price is equal to the current price of the stock), this inequality says that call prices should be greater than put prices.

For our six cases with negative stubs, three had exchange-traded American options within the relevant time frame: Xpedior, Palm, and Stratos. We used weekly share prices and weekly options prices, as of 4:00 P.M. eastern time on Friday.

Table 6 shows an example from the first week of trading in Palm's options (occurring more than two weeks after the IPO) using options that are closest to being at-the-money. Options on Palm display massive violations of put-call parity and violate the weaker inequality as well. Instead of observing at-the-money call prices that are greater than put prices, we find that puts were about twice as expensive as calls. We also calculate the implied price of synthetic securities. For example, on March 17, one can create a synthetic short position in Palm by buying a November put (at the ask price), writing a November call (at the bid price), and borrowing dollars. Both the synthetic short and the actual short position, if held until November, give the same payoff of the negative of the price of Palm in November. These calculations are done using the assumption that one can borrow from March to November at the six-month London Interbank offer rate (LIBOR). On March 16

⁷ This effect was present in our sample, since in the three cases with negative stubs, when exchange-traded options were introduced, all three had sizable increases in the stub value. In all three cases, the subsidiary fell on the day on which options started trading.

TABLE 6
PALM OPTIONS ON MARCH 17, 2000
A. OPTIONS PRICES

	CALL		PUT		SYNTHETIC SHORT	PERCENTAGE DEVIATION	SYNTHETIC LONG	PERCENTAGE DEVIATION
	Bid	Ask	Bid	Ask				
May 55	5.75	7.25	10.625	12.625	47.55	-14	51.05	-8
August 55	9.25	10.75	17.25	19.25	43.57	-21	47.07	-15
November 55	10	11.5	21.625	23.625	39.12	-29	42.62	-23
B. OTHER PRICES								
LIBOR:								
3-month		6.21						
6-month		6.41						
Stock prices:								
Palm		55.25						
3Com		69						

NOTE.—May options expire May 20, 2000; August options expire August 19, 2000; and November options expire November 18, 2000. A synthetic short position buys a put (at the ask price), sells a call (at the bid price), and borrows the present value of the strike price. A synthetic long position sells a put (at the bid price), buys a call (at the ask price), and lends the present value of the strike price. We discount May cash flows by three-month LIBOR and August and November cash flows by six-month LIBOR. Source of options price data: Chicago Board Options Exchange. Source of LIBOR: Datastream

the price of the synthetic short was about \$39.12, far below the actual trading price of Palm of \$55.25. This constellation of prices is a significant violation of the law of one price since the synthetic security is worth 29 percent less than the actual security. May and August options also showed substantial, though smaller, violations of put-call parity.

The synthetic shorts at different horizons in table 6 can be used to calculate the implied holding cost of borrowing Palm's shares. For an investor who is indifferent to shorting actual Palm shares from March until May and creating a synthetic short, the holding costs must be 14 percent over two months, or about 119 percent at an annual rate. For an investor planning to short for eight months, until November, the holding costs must be 29 percent, or 147 percent at an annual rate. Thus the options prices suggest either that shorting Palm was incredibly expensive or that there was a large excess demand for borrowing Palm shares, a demand that the market could not meet for some institutional reasons.

Since the evidence from D'Avolio (2000) indicates a much lower, 35 percent, shorting cost for Palm during this period, it is clear that there must be other risks and costs associated with shorting Palm. First, there is the cost of actually finding shares to borrow. Second, as discussed in Liu and Longstaff (2000) and Mitchell et al. (2002), short sellers are required to post additional collateral if the price of Palm rises. Third, as discussed in Mitchell et al., there is "buy-in" risk, the fact that the Palm lender has the right to recall his loan at any time. If the Palm

lender decides to sell his shares after they have risen in price, the short sellers may be forced to close their position at a loss if they are unable to find other shares to borrow. Fourth, even if the loan is not recalled, the cost of shorting could increase if the rebate changes.

We now have three different market estimates of Palm's value: the embedded value reflected in 3Com's share price, the value reflected in options prices, and the actual share price. The options market and the shareholders in 3Com seemed to agree: Palm was worth far less than its market price. The direction of the deviation from the law of one price is consistent with the difficulty of shorting Palm. To profit from the difference between the synthetic security and the underlying security, one would need to short Palm and buy the synthetic long. The price of the synthetic short reflects the high demand for borrowing Palm stock and the low supply. Similarly, Figlewski and Webb (1993) find that, in general, stocks with high short interest have puts that are more expensive relative to calls (although they look at implied volatilities instead of put-call parity).

Again, although the prices here are consistent with very high shorting costs, one can turn the inequality around and ask why anyone would ever buy Palm (without lending it). On March 17 one can create a synthetic long Palm by buying a call and selling a put, and this synthetic long is 23 percent cheaper than buying an actual share of Palm and holding it until November.⁸ Arguments about the risk that the planned spin-off may not occur are irrelevant to the synthetic long constructed using options. Why are investors who buy Palm shares directly willing to pay much more than they could pay using the options market? One plausible explanation is that the type of investor buying Palm is ignorant about the options market and unaware of the cheaper alternative.⁹

One can use the synthetic short price of Palm to create a synthetic stub value. On March 17, 2000, the actual stub value for Palm was $-\$16.26$ per share. The synthetic stub for Palm, constructed using the synthetic short price implied in six-month at-the-money options, was positive at $\$1.56$. Although this value seems low (i.e., less than the cash

⁸ Of course, the put-call parity formula holds only for stocks paying no dividends. One benefit of owning Palm is that it yields a "dividend" from lending it out to short sellers. As before, however, someone is holding all the Palm stock without lending it out; this owner would be better off owning the synthetic short.

⁹ In the model of Duffie, Garleanu, and Pedersen (2001), all investors seek to lend the stock out in order to reap lending income. Since the securities lending market works sluggishly, although everyone is trying to lend, not all succeed, and at any one time 100 percent of the shares are owned by someone. It is not clear whether this explanation is quantitatively plausible in the case of Palm, since the reported lending income is substantially less than the amount necessary to make buyers indifferent between owning actual shares and synthetic shares.

3Com held), it is at least positive and thus no longer so close to a pure arbitrage opportunity.

We have seen earlier that the actual stubs became less negative over time and eventually turned positive. In figure 5 we display the time series of the actual stubs along with the synthetic stubs for the time period up to the distribution date (constructing synthetic stubs using options that are closest to six months and at-the-money). The solid line, the actual stub, goes from strongly negative at the beginning to positive \$10 a share. The dotted line, the synthetic stub, is positive in all but one week. By the distribution date, the difference between the two lines is close to zero, roughly consistent with put-call parity. The pattern shows that options prices adjust to virtually eliminate profitable trading opportunities. Put differently, the implied cost of shorting falls as the desirability of shorting falls.

Figure 6 shows the case for Stratos. The pattern is similar; again, there is a single week in which the synthetic stub is negative at the beginning, and the synthetic stub stays around \$5 per share, correctly forecasting the eventual free-standing price of the parent. As the stub becomes less negative, the gap between the actual and synthetic stub narrows. Thus Stratos also supports the idea that the high cost of shorting allows the new subsidiary to be overpriced.

Our third case with exchange-traded options is Xpedior. Unfortunately, Xpedior is a marginal case, and it produces a stub that is strongly negative for only one week when options are trading. When we examine the difference between actual and synthetic prices (not shown in a figure), Xpedior does not seem to display a high cost of shorting, although we have little power since the actual stub is so marginally negative.

It is intriguing that in figures 5 and 6 there are some negative synthetic stub observations that seem to be exploitable opportunities. We can report that these opportunities truly were exploitable and reflected actual prices, since one of the authors (Lamont) traded on these opportunities and made profits. We suspect that these opportunities were quite limited in size, however, since individual equity options are illiquid, with low volume, low open interest, and high price impact. If arbitrageurs had attempted to buy a few million dollars worth of puts, it seems likely that the price of puts would have risen to eliminate profitable opportunities. However, it still remains a puzzle why arbitrageurs did not buy until the prices adjusted.

In table 7, we regress the violation of put-call parity (the deviation of the synthetic stub) on the actual stub for Palm and Stratos. For Palm, the synthetic stub deviation moves strongly with the actual stub, and even with just 19 weekly observations, we can reject the hypothesis that the two do not move together. The R^2 is a whopping .96, suggesting

TABLE 7
REGRESSION OF SYNTHETIC STUB DEVIATION
ON ACTUAL STUB

	Palm	Stratos
Constant	-8.15 (.24)	-5.95 (.50)
S_i	.50 (.02)	.83 (.08)
Observations	19	42
R^2	.96	.71

NOTE.—The dependent variable is $S_i - S_s$, the deviation between the actual stub and the synthetic stub, expressed in dollars per parent share. The actual stub, $S_i = P_i^p - x_i P_i^s$, uses actual prices of the shares. The synthetic stub, $S_i = P_i^p - x_i P_i^s$, uses the actual price of parent shares and the synthetic short price of subsidiary shares. The synthetic short price, $-P_i^s$, is constructed by selling a six-month at-the-money call at the bid prices, buying a six-month at-the-money put at the ask prices, and borrowing the present value of the exercise price at the six-month LIBOR rate. The regression for Palm uses 19 weekly observations as of Friday March 17, 2000–July 21, 2000; the regression for Stratos uses 42 weekly observations as of Friday July 14, 2000–April 27, 2001.

that violations of put-call parity are strongly related to apparent near-arbitrage opportunities. For Stratos, the R^2 is lower at .70, but again we can easily reject the hypothesis that the stub and the deviation of actual from synthetic are unrelated.

Are these violations of put-call parity unusual? Most empirical studies of options prices have found that put-call parity basically holds, with small or fleeting violations due perhaps to trading costs or asynchronous price data (Klemkosky and Resnick 1979; Bodurtha and Courtadon 1986). One might wonder whether put-call parity generally holds using data from our sample period and using our sources and methods. Although a thorough investigation of put-call parity for all equity options is beyond the scope of this paper, we did do a brief check as follows. We picked a random date, October 10, 2000, and compared the synthetic short on Stratos with those of other options. Stratos options started trading on the Chicago Board Options Exchange (CBOE) on July 12, 2000. We looked at 28 other firms in which options were initially listed on the CBOE between June 11 and July 12, 2000. Most of these firms were, like Stratos, recent technology IPOs. We omitted firms paying dividends or firms with a stock price below \$10 a share. On October 10, the stub value for Stratos was -\$1.66 a share, and the synthetic short price constructed using six-month options was 24 percent below the actual price of Stratos (similar to the deviation seen for Palm in table 6), or \$5.89 below the actual price per share. For the 28 other firms, the average synthetic short price was only 3 percent below the actual price, or 87 cents per share, easily explainable with bid/ask spreads on options. The maximum deviation was 8 percent below the actual price,

only a third of the deviation observed for Palm and Stratos. On the basis of this evidence, the Palm and Stratos cases appear to present unusually large violations of put-call parity.

To conclude, in the case of Palm and Stratos, we have strong evidence from options markets confirming that the new issues are overpriced, and no one should buy them (at least without lending them out, which not everyone can do in equilibrium) because cheaper alternatives are available. Although shares in the parent are not perfect substitutes for shares in the subsidiary (because of the risk of spin-off), the synthetic shares are virtually identical. Although not an exploitable arbitrage opportunity, this is a case of blatant mispricing.

V. What Causes Mispricing?

We hope to have convinced even the most jaded reader that the cases we are studying are clear violations of the law of one price. Given that arbitrage cannot correct the mispricing, why would anyone buy the overpriced security? Why are some investors willing to buy shares in Palm when there are cheaper alternatives available in the market, either by buying the parent or by buying Palm synthetically in the options market? In this section we investigate this question, first by asking a simple question: Who buys the expensive subsidiary shares, and how long do they hold them? We then look at IPO day returns for evidence on how these investors affect prices of the parent.

A. *Investor Characteristics*

Columns 1 and 2 of table 8 display volume data for both parents and subsidiaries in our six cases with negative stubs. We show turnover for the first 20 days of trading, defined as average daily volume divided by shares outstanding (for parents) or by total shares sold to the public (for the IPO). The turnover measure does not include the first day of trading itself. All 12 stocks trade on NASDAQ. Since NASDAQ is a dealer market, reported volume includes dealer trades, and the turnover caused by trades between actual investors is approximately half the turnover reported in table 8.

The first thing to note is that subsidiaries have turnover that is more than five times that of parent turnover, with 37.8 percent of all tradable shares turning over *per day*. Higher turnover means that subsidiary shareholders have lower holding periods, and thus shorter horizons, compared to parent shareholders. Nondealer shareholders of UBID, for example, had an average horizon of two trading days, since turnover was more than 100 percent (implying 50 percent turnover, with dealer trades excluded).

TABLE 8
VOLUME, LIQUIDITY, AND INSTITUTIONAL OWNERSHIP

	TURNOVER		BID/ASK SPREAD		INSTITUTIONAL OWNERSHIP	
	Parent (1)	Subsidiary (2)	Parent (3)	Subsidiary (4)	Parent (5)	Subsidiary (6)
Creative/UBID	23.98	106.47	.69	.93	17.71	10.38
HNC/Retek	3.68	22.19	.32	.26	96.38	72.28
Daisytek/ PFSWeb	2.42	25.53	.62	.81	71.88	69.95
Metamor/ Xpedior	2.13	11.79	.42	.49	53.06	35.96
3Com/Palm	4.54	19.18	.09	.14	52.22	46.01
Methode/ Stratos	2.63	41.67	.42	.20	69.47	36.63
Average	6.56	37.80	.43	.47	60.12	45.20
Difference, parent vs. subsidiary		31.24		.04		-14.92
t-statistic		2.83		.62		-3.06

NOTE.—Turnover is daily volume as a percentage of parent shares outstanding or subsidiary shares trading. Subsidiary shares trading are shares sold to the public in the IPO. Volume is average daily volume from the first 20 trading days after the IPO date (not including the first day of trading). The shares outstanding of the parent are taken from CRSP, and the shares issued in the IPO are taken from company SEC filings. Bid/ask spread is the average percentage of prices from the first 20 trading days after the IPO date (not including the first day of trading). Institutional ownership, from 13F filings to the SEC (via Securities Data Corp.), pertains to the first quarterly filing after the IPO. Institutional ownership refers to a percentage of parent shares outstanding or subsidiary shares trading.

These turnover figures suggest that the subsidiaries may have been more liquid than the parents. If investors value liquidity, then more liquid securities should have higher value and should have higher turnover. To investigate this possibility, columns 3 and 4 of table 8 report bid/ask spreads as a percentage of price for the first 20 days of trading. Contrary to the hypothesis of greater liquidity, there is no significant difference in bid/ask spread for the parents and subsidiaries.

Another interpretation of high volume is that it is consistent with the greater fool theory, where investors buy the subsidiary knowing that it is overpriced but hoping it will rise even higher. If they are holding the stock for only a few days, they might not care that they can obtain a cheaper version with identical payoff six months from now (see also Cochrane 2002). Formalizations of this idea include Harrison and Kreps (1978) and De Long et al. (1990). Harrison and Kreps have a model in which agents have different beliefs and act rationally conditional on these beliefs, but short-sale constraints mean that only optimistic investors hold the stock. The model has the remarkable property that stock prices can be above the valuations of the most optimistic investors. In some cases, all investors agree that the stock is overpriced, yet some are still willing to hold it. The reason is that they all think they are following a dynamic strategy that allows them to cash out when the stock gets

really overpriced. An essential part of this story is the dynamic trading strategy generating high volume and low holding periods.

Columns 5 and 6 of table 8 also show institutional ownership for parents and subsidiaries using data from quarterly 13F filings, reflecting holdings by institutional investment managers having equity assets under management of \$100 million or more. In the first quarter after the IPO, institutional ownership is 15 percent higher for parents than for subsidiaries (this difference is understated because of the heavy short interest in subsidiaries).¹⁰

One potential explanation for the mispricing involves restrictions on what institutions are allowed to hold. For example, Froot and Dabora (1999) show that Royal Dutch and Shell (two stocks representing the same firm) seem mispriced relative to each other. In recent years, the stock that is part of the Standard & Poor's 500 (Royal Dutch) trades at a premium to the stock that is not (Shell), possibly reflecting the fact that index funds are forced to buy the more expensive stock and cannot substitute the cheaper one. Similarly, one money manager told us (discussing stub situations in general) that although he was well aware that a particular subsidiary was overpriced relative to the parent, he could not buy the cheaper parent instead of the subsidiary because he ran a growth fund, and the cheaper stock was, by definition, value! However, table 8 suggests that such institutional explanations are unlikely to explain the overpricing since most owners are individuals.

The information in table 8 also helps explain why the supply of lendable shares to short was so sluggish. First, high turnover impedes securities lending because when a share lender sells his shares, the share borrower is obliged to return the shares and must find a new lender. Second, shares held by individual investors are less likely to be lent than shares held by institutions.

In summary, table 8 shows that subsidiaries had very high turnover but not high liquidity and had low institutional ownership. This evidence is perfectly consistent with the view that irrational or ignorant investors drove up the price of the subsidiary shares and limits to arbitrage prevented rational investors from correcting this mispricing. We next turn to evidence from IPO day returns for additional evidence.

¹⁰ We report institutional ownership as a percentage of parent shares outstanding or subsidiary shares trading. For example, Palm sold 26.5 million shares in the IPO on March 2, 2000, had 5.1 million shares in short interest as of March 15, and had institutional ownership of 12.1 million shares at the end of March. Although 26.5 million shares were issued, 31.6 million shares were owned by somebody, thanks to short sellers who borrowed shares and sold them. Thus institutions held 46 percent of the shares issued, but only 38 percent of all the ownable shares.

TABLE 9
IPO DAY RETURNS FOR ENTIRE CARVE-OUT SAMPLE

	SUBSIDIARY			PARENT		
	Offer Price	Closing Price	Percentage Change	Pre-IPO Price	Closing Price	Percentage Change
HP/Agilent	30.00	42.75	43	78.00	94.31	21
Odetics/ATL	11.00	11.88	8	19.63	18.25	-7
Eaton/Axcelis	22.00	23.94	9	69.50	69.50	0
Viacom/Blockbuster	15.00	15.00	0	40.56	39.94	-2
Cabot Corp/Cabot Micro	20.00	24.88	24	29.50	28.00	-5
Cincinnati Bell/ Convergys	15.00	16.63	11	29.75	28.69	-4
GM/Delphi	17.00	18.63	10	87.06	85.94	-1
Deluxe/Efunds	13.00	12.00	-8	23.88	23.31	-2
AT&T/Lucent	27.00	30.63	13	64.13	62.88	-2
Santa Fe/Monterey	14.50	16.50	14	14.75	15.00	2
Sea Containers/ Orient Express	19.00	19.75	4	28.13	26.25	-7
HNC/Retek	15.00	32.56	117	61.00	60.88	0
Tridex/TransAct	8.50	8.75	3	10.44	10.63	2
Metamor/Xpedior	19.00	26.00	37	33.19	29.00	-13
Average for 14 subsidiaries with positive stub on first day			20			-1
3Com/Palm	38.00	95.06	150	104.13	81.81	-21
Daisytek/PFSWeb	17.00	44.13	160	22.63	21.94	-3
Methode/Stratos	21.00	34.13	63	43.94	41.88	-5
Creative/UBID	15.00	48.00	220	35.25	26.25	-26
Average for four subsidiaries with negative stub on first day			148			-14
t-statistic for difference in means, 14 carve- outs vs. four carve-outs			5.69			2.61

NOTE.—Daily closing prices are taken from CRSP. Pre-IPO price is the price of the parent on the day previous to the IPO.

B. IPO Day Returns

Hand and Skantz (1998), looking at carve-outs generally, provide evidence that irrational investors can affect carve-out pricing. As documented in Schipper and Smith (1986) and Allen and McConnell (1998), when announcing the carve-out, parents earn excess announcement returns of around 2 percent. Hand and Skantz show that on the IPO date itself, parents have excess returns of -2 percent. One explanation is that optimistic investors who desire to hold the subsidiary drive up the price of the parent on the announcement days and then dump the parent in favor of the subsidiary on the IPO day.

Table 9 looks at evidence for segmentation in our sample from IPO day returns. It compares IPO day returns for the 14 subsidiaries that had positive stubs on the IPO date and the four subsidiaries with negative stubs (for Xpedior and Retek the stubs became negative after only a few days of trading). Table 9 shows that subsidiaries resulting in negative stubs had much higher IPO returns than other subsidiaries, where the

returns are offer price to closing price for the new subsidiary. This difference is unsurprising since one way to get negative stubs is to have a high price of the subsidiary.

Another way to get a negative stub is to have a low price of the parent. Table 9 also shows that the prices of parents in negative stub situations fell 14 percent from the day before the IPO to the close on the IPO day. For the 14 cases with positive stubs on the IPO date, the parents fell an average of 1 percent. The differences between the positive stub and negative stub IPOs are large and statistically significant for both parent returns and subsidiary returns (the statistical significance does not change if one categorizes Xpedior and Retek, which had negative stubs in the next few days, in the second group).

The large decline in parent prices in negative stub situations is surprising since the parents own so much of the new issue. One might think that when the subsidiary does unexpectedly well on the issue date, the parent would benefit as the value of its holdings increases. For example, prior to the issue, Palm's underwriters had originally estimated the offering price to be \$14–\$16 per share. After gauging investor demand, they increased the estimated offering price to \$30–\$32. Finally, the night before the offer, they chose \$38 as the final issuing price. On the first day of trading, Palm immediately went to \$145 and later rose to as high as \$165, before ending the day at \$95.06 a share. Thus the very high subsidiary return seems likely to have been a surprise, making the drop in the price of 3Com that day mystifying.¹¹

These patterns are all consistent with irrational investors. Prior to the IPO, irrational optimists who desire to own Palm have to hold 3Com instead. 3Com trades in the optimistic segment of the market. Once the IPO occurs, these optimists buy Palm directly (ignoring the cheaper alternative of holding 3Com). 3Com now trades in the more rational segment of the market, and its price falls to the rational price.

VI. Conclusion

One of us used to have a colleague who, when teaching the basic finance course to impressionable young first-year master of business administration students, would shout the name of a well-known game show as a key conclusion of efficient markets: The Price Is Right! He would offer little empirical support for this claim, but could rest assured that it was

¹¹ More generally, Bergstresser and Karlan (2000) examine cross-corporate equity holdings similar to the ones considered here (but without the terminal date) and find that parent firm stock prices underreact to changes in the value of their holdings. Similarly, closed-end funds trading in the United States but holding foreign securities have prices that do not always react properly to foreign market movements (see Klibanoff, Lamont, and Wizman 1998).

a claim that was hard to disprove. The trick to testing the “price is right” hypothesis is to find unambiguous relative price comparisons, such as closed-end funds.

The negative stubs in this paper are in a similar category, though the mispricing appears to be even more blatant. In contrast to closed-end funds, where arguments about agency costs by the fund managers, tax liabilities, and bad estimates of net asset value can cloud the picture, in this case any investor who can multiply by 1.5 should be able to tell that Palm is overpriced relative to 3Com. The evidence from options markets shows that these stocks were unambiguously overpriced, and it is difficult to explain why in equilibrium anyone would own these shares. The mispricing persisted because of the sluggish functioning of the shorting market

There are two key findings of this paper that need to be understood as a package. First, we observe gross violations of the law of one price. Second, they do *not* present exploitable arbitrage opportunities because of the costs of shorting the subsidiary. In other words, the no free lunch component of the efficient market hypothesis is intact, but the price equals intrinsic value component takes another beating.

Still, it is possible to argue that we have only six cases here that collectively represent a tiny portion of the U.S. equity market. Maybe everything else is just fine. Why should we be concerned? Put another way, are these cases of blatant mispricing the tip of a much bigger iceberg or the entire iceberg? In one respect, our overpriced stocks are clearly different from most stocks. They were difficult or expensive to borrow because the supply of lendable shares did not quickly respond to the mispricing. In contrast, most stocks and particularly large-cap stocks are easy to borrow. Reed (2001) and D’Avolio (2002) show that few stocks are expensive to short, and Figlewski and Webb (1993) report that average short interest as a percentage of outstanding shares is only 0.2 percent. Although Ofek and Richardson (2001) report that Internet stocks had higher average short interest and were more expensive to short than non-Internet stocks in the period we study, the average difference in cost was only 1 percent per year. So perhaps it is only the rare cases in which shorting is very expensive that lead to mispricing. That is the rosy interpretation of our findings.

There is another interpretation, however, that is less rosy but more plausible. We think that a sensible reading of our evidence should cast doubt on the claim that market prices reflect rational valuations because the cases we have studied should be ones that are particularly easy for the market to get right. Suppose we consider the possibility that Internet stocks were priced much too high around 1998–2000. The standard efficient markets reaction to such claims is to say that this cannot happen. If irrational investors bid up prices too high, arbitrageurs will step

in to sell the shares short and, in so doing, will drive the prices back down to rational valuations. The lesson to be learned from this paper is that arbitrage does not always enforce rational pricing. In the case of Palm, arbitrageurs faced little risk but could not find enough shares of Palm to satiate the demands of irrational investors. We have identified cases in which arbitrageurs are *unable* to arbitrage relative mispricing. More generally, there can be cases of mispricing in which arbitrageurs are *unwilling* to establish positions because of fundamental risk or noise trader risk. Many investors thought that Internet stocks were overpriced during the mania, but only a small minority were willing to take a short position, and these short sellers were not enough to drive prices down to rational valuations. Further, many institutions either are not permitted to sell short or simply choose not to do so for various reasons. Almazan et al. (2001) find that only about 30 percent of mutual funds are allowed to sell short, and only 2 percent actually do sell short.

Limits of arbitrage can create market segmentation. If irrational investors are willing to buy Palm at an unrealistically high price and rational but risk-averse investors are unwilling or unable to sell enough shares short, then two inconsistent prices can coexist. The same argument can apply to any apparent mispricing, from closed-end fund discounts and premia to differences in returns between value stocks and growth stocks. The traditional view is that a stock with a low expected return must have low risk. The examples given here suggest an alternative possibility, namely that the investors who buy apparently expensive stocks are just making a mistake.

The conclusion we draw is that there is one law of economics that does still hold: the law of supply and demand. Prices are set so that the number of shares demanded equals the number of shares supplied. In the case of Palm, the supply of shares could not rise to meet demand because of the sluggish response of lendable shares to short. Similarly, if optimists are willing to bid up the shares of some faddish stocks and not enough courageous investors are willing to meet that demand by selling short, then optimists will set the price.

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